Quarkus Essentials Part Two

☕🚀productivity and efficiency

airhacks.live

"It's not work if you like it"
…so I never worked. #java

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#193 Working in the Shadows ...for Quarkus

#76 Quarkus Developer Experience

#64 Quarkus 1.0 and SpringBoot

[episode link] Listen on Apple Podcasts LISTEN ON Spotify Listen on Google Podcasts [RSS]

#52 The First Line of Quarkus

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An airhacks.fm conversation with Emmanuel Bernard (@emmanuelbernard) about:

> Amstrad PC 1512, the first PC computer ever made by Amstrad, two floppies are better than a hard drive, deleting double dots, C/PM OS, BIOS as cheating detection, creating snake game in BASIC, playing with Turbo Pascal, from GO TOs over loops to procedures, objects, aspects to functional programming, exploring Mandrake Linux, Tibco Messaging and C++, killing yourself with casting, the C discipline checker - an enforced linter, 120 errors caused by

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with the time machine, “100 episodes ago segment”

…any questions left?

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NEW online, live virtual workshops

Continuous coding, explaining, interacting and sharing with Adam Bien

Live, Virtual Online Workshops, Winter 2025:

FULLY BOOKED The 50x Java Dev: Crafting Elegant Code with LLMs, 16 December 2025

...or how to write maintainable code with LLMs fast

FULLY BOOKED Architecting Applications with MCP, A2A and AI Agents, 18 December 2025

...or how to write agentic applications

Live, Virtual Online Workshops, Winter 2026:

Effective Java Evolution: Incremental Development and Continuous Improvement with LLMs, 26 February 2026

...or how to incrementally write maintainable code with LLMs

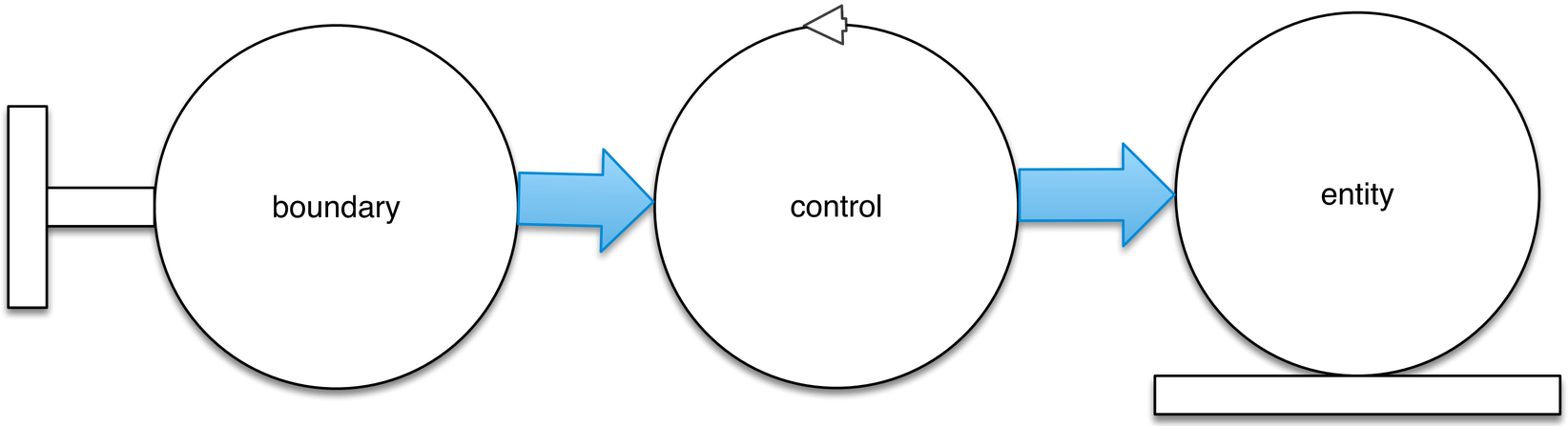
Tickets are also available from: airhacks.eventbrite.com and meetup.com/airhacks

the ultimate internal structure

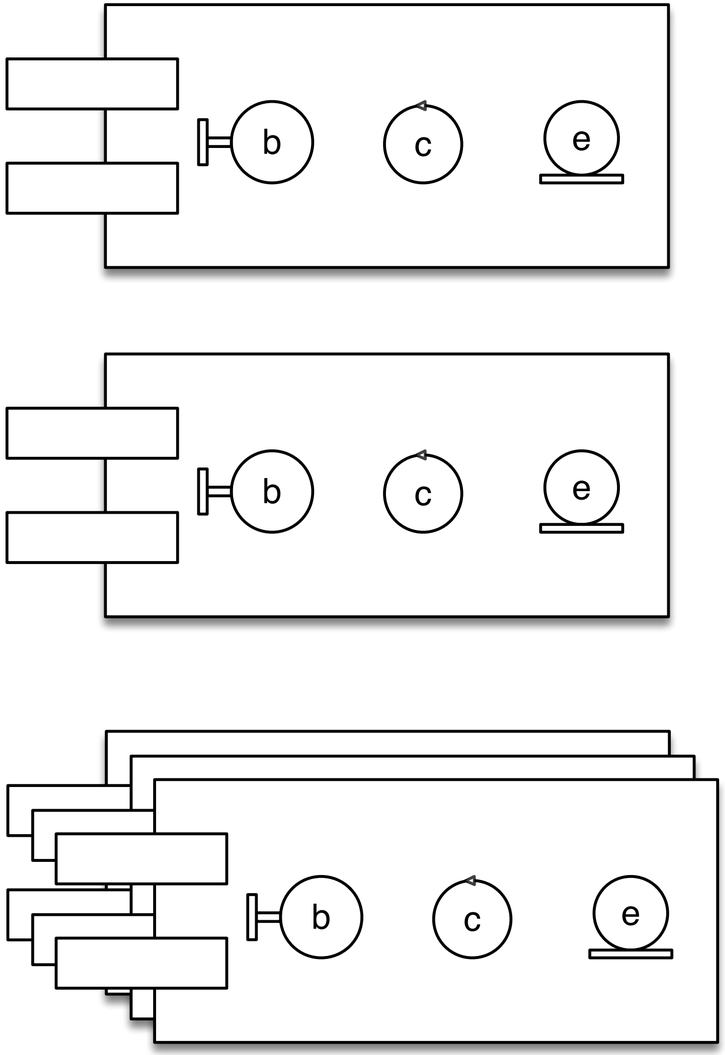
Boundary Control Entity

- ancient pattern / icons available for all tools
- top level packages are named after business responsibilities
- boundary: the perimeter / the facade
- control: procedural logic
- entity: data and domain logic
- works with frontends and backends

Boundary Control Entity



Boundary Control Entity



https://bce.design

BCE Design

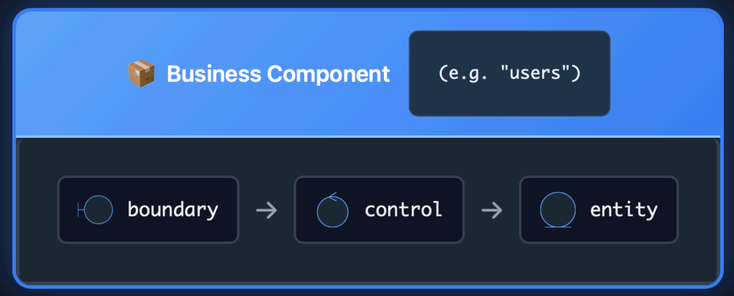
Overview | Business Components | Boundary | Control | Entity | Interactions | Benefits | References | Architectural Styles ▼

Boundary Control Entity Architecture

Focus on building understandable applications with a strong emphasis on domain logic

Overview

The Boundary-Control-Entity (BCE/ECB) pattern is a software architecture pattern that organizes code into Business Components. A Business Component is a package or namespace comprising three distinct layers, each with specific responsibilities. Business components adhere to the principles of maximal cohesion and minimal coupling, and are named after their domain responsibilities.



test

test

- System Tests first
- Torture / Stress Tests next
- App and System Tests in separate maven projects
- save time with MicroProfile REST client
- JAX-RS resources -> MP REST Client interfaces

test

```
@Path("hello")
@ApplicationScoped
public class GreetingResource {

    @Inject
    Greeter greeter;

    @GET
    @Produces(MediaType.TEXT_PLAIN)
    public String hello() {
        return this.greeter.greetings();
    }
```

test

```
import javax.ws.rs.GET;
import javax.ws.rs.Path;
import javax.ws.rs.Produces;
import javax.ws.rs.core.MediaType;

import org.eclipse.microprofile.rest.client.inject.RegisterRestClient;

@Path("hello")
@RegisterRestClient(configKey = "base_uri")
public interface GreetingsResource {

    @GET
    @Produces(MediaType.TEXT_PLAIN)
    Response content();

}
```

test

```java
@QuarkusTest
public class GreetingsResourceIT {

    @Inject
    @RestClient
    GreetingsResource resource;

    @Inject
    @ConfigProperty(name = "base_uri/mp-rest/url")
    String baseURI;

    @Test
    public void hello() {
        var message = this.resource.content();
        assertNotNull(message);
        System.out.println(message);
    }

}
```

test

- Rest Client For Microprofile: interface reuse
- high-level tests are increasing productivity (reuse possible)
- CI/CD pipeline ships with stages

references

- “Ten Years After: From Java EE 6 to Quarkus and AWS Lambda”: http://youtube.com/@bienadam/
- “Hey Enterprise EJB Developers Now Is The Time To Go Serverless”: http://youtube.com/@bienadam/
- “Past, Present and Future: Serverside Java on Premise and in the Clouds”: http://youtube.com/@bienadam/
- https://github.com/AdamBien/x-ray (~15 years old code)
- https://github.com/AdamBien/lightfish (~15 years old code)

examples / resources

- https://bce.design
- https://github.com/AdamBien/quarkus-microprofile
- https://github.com/adambien/ebank
- https://www.youtube.com/bienadam/shorts
- https://github.com/AdamBien/microprofile.training
- https://constructions.cloud
- Questions? 👉 https://airhacks.tv

Thank YOU!